(i)

$$\because \text{Sum of all probability is 1}$$

,

$$\therefore \sum_{x=1}^{n}\sum_{y=1}^{n} P(X=x, Y=y) = 1$$

$$\implies \sum_{x=1}^{n}\sum_{y=1}^{n} k(x+y) = 1$$

$$\implies k\left(\sum_{x=1}^{n}\sum_{y=1}^{n} x + \sum_{x=1}^{n}\sum_{y=1}^{n} y\right) = 1$$

$$\implies k\left(n\sum_{x=1}^{n} x + n\sum_{y=1}^{n} y\right) = 1$$

$$\implies k\left(n \cdot \frac{n(n+1)}{2} + n \cdot \frac{n(n+1)}{2}\right) = 1$$

$$\implies \boxed{k = \frac{1}{n^2(n+1)}}$$

$$P(X=x) = \sum_{y=1}^{n} P(X=x, Y=y) = \sum_{y=1}^{n} k(x+y)$$

$$= k\left(\sum_{y=1}^{n} x + \sum_{y=1}^{n} y\right)$$

$$= k\left(nx + \frac{n(n+1)}{2}\right)$$

$$= \frac{1}{n^2(n+1)}\left(nx + \frac{n(n+1)}{2}\right)$$

$$= \frac{2x+n+1}{2n(n+1)}$$

$$\implies \boxed{P(X=x) = \frac{n+1+2x}{2n(n+1)}}$$

By symmetry, $P(Y=y) = \dfrac{n+1+2y}{2n(n+1)}$

If $P(X=x, Y=y) = P(X=x) \times P(Y=y)$, then X and Y are independent.
When $x=1, y=1$:

$$P(X=1) \times P(Y=1) = \left(\frac{3+n}{2n(n+1)}\right)^2 \qquad P(X=x, Y=y) = \frac{x+y}{n^2(n+1)} = \frac{2}{n^2(n+1)}$$

$$P(X=1)\times P(Y=1)-P(X=1,Y=1)=\left(\frac{n+3}{2n(n+1)}\right)^2-\frac{2}{n^2(n+1)}$$

$$=\frac{(n+3)^2-8(n+1)}{4n^2(n+1)^2}$$

$$=\frac{n^2-2n+1}{4n^2(n+1)^2}$$

$$=\frac{(n-1)^2}{4n^2(n+1)^2}>0 \qquad (n\geq 2)$$

$$\therefore P(X=1)\times P(Y=1)\neq P(X=1,Y=1)$$

(ii)

$$\mathrm{Cov}(X,Y)=E(XY)-E(X)E(Y)$$

$$E(X)=E(Y)=\sum_{x=1}^{n}xP(X=x)$$

$$=\sum_{x=1}^{n}x\cdot\frac{n+1+2x}{2n(n+1)}$$

$$=\frac{1}{2n(n+1)}\left((n+1)\sum_{x=1}^{n}x+2\sum_{x=1}^{n}x^2\right)$$

$$\because \sum_{x=1}^{n}x=\frac{1}{2}n(n+1) \quad \text{and} \quad \sum_{x=1}^{n}x^2=\frac{1}{6}n(n+1)(2n+1)$$

$$\therefore E(X)=\frac{1}{2n(n+1)}\left((n+1)\cdot\frac{n(n+1)}{2}+2\cdot\frac{n(n+1)(2n+1)}{6}\right)$$

$$=\frac{1}{2n(n+1)}\left(\frac{n(n+1)^2}{2}+\frac{n(n+1)(2n+1)}{3}\right)$$

$$=\frac{1}{2}\left(\frac{n+1}{2}+\frac{2n+1}{3}\right)$$

$$=\frac{7n+5}{12}$$

$$
\begin{aligned}
E(XY) &= \sum_{x=1}^{n}\sum_{y=1}^{n} xy\,P(X=x,Y=y) \\
&= \sum_{x=1}^{n}\sum_{y=1}^{n} xy\cdot k(x+y) \\
&= k\sum_{x=1}^{n}\sum_{y=1}^{n}(x^2y+xy^2) \\
&= k\left(\sum_{x=1}^{n}\sum_{y=1}^{n}x^2y+\sum_{x=1}^{n}\sum_{y=1}^{n}xy^2\right) \\
&= k\left(2\cdot\sum_{x=1}^{n}\sum_{y=1}^{n}x^2y\right) \\
&= k\left(2\cdot\left(\sum_{x=1}^{n}x^2\right)\left(\sum_{y=1}^{n}y\right)\right) \\
&= k\left(2\cdot\frac{n(n+1)(2n+1)}{6}\cdot\frac{n(n+1)}{2}\right) \\
&= \frac{1}{6}(n+1)(2n+1)
\end{aligned}
$$

$$
\begin{aligned}
\implies \operatorname{Cov}(X,Y) &= \frac{1}{6}(n+1)(2n+1)-\left(\frac{7n+5}{12}\right)^2 \\
&= \frac{24(n+1)(2n+1)-(7n+5)^2}{144} \\
&= \frac{48n^2+72n+24-49n^2-70n-25}{144} \\
&= \frac{-n^2+2n-1}{144} \\
&= -\frac{(n-1)^2}{144}
\end{aligned}
$$

$$
\because (n-1)^2 > 0
$$

$$
\therefore \boxed{\operatorname{Cov}(X,Y) = -\frac{(n-1)^2}{144} < 0}
$$

This solution was submitted by **Jon**

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